FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS

FULL TEST – V
(Paper-2)

Q. No.	PHYSICS	Q. No.	CHEMISTRY	Q. No.	MATHEMATICS
1.	**C**	24.	**D**	47.	**C**
2.	**B**	25.	**A**	48.	**B**
3.	**B**	26.	**A**	49.	**A**
4.	**C, D**	27.	**B, C**	50.	**A, B, C, D**
5.	**B, C**	28.	**B, C, D**	51.	**A, B, D**
6.	**A, B, D**	29.	**B, D**	52.	**A, C**
7.	**C, D**	30.	**B, C**	53.	**B, D**
8.	**A, B, C, D**	31.	**A, B, C, D**	54.	**B, C**
9.	**B**	32.	**C**	55.	**C**
10.	**B**	33.	**A**	56.	**B**
11.	**2**	34.	**3**	57.	**5**
12.	**5**	35.	**7**	58.	**6**
13.	**8**	36.	**6**	59.	**4**
14.	**1**	37.	**5**	60.	**3**
15.	**3**	38.	**1**	61.	**8**
16.	**4**	39.	**3**	62.	**2**
17.	**1**	40.	**8**	63.	**3**
18.	**4**	41.	**4**	64.	**3**
19.	**5**	42.	**2**	65.	**5**
20.	**6**	43.	**3**	66.	**4**
21.	**00003.00**	44.	**00080.00**	67.	**00001.12**
22.	**00003.00**	45.	**00017.92**	68.	**00000.75**
23.	**01000.00**	46.	**00049.50**	69.	**00000.75**

Physics

PART – I

SECTION – A

1. **C**

(C) Direction of friction does not change.

(D) Friction is constant.

2. **B**

$\sin\theta_C = \mu_2 / \mu_1$

$1 - \cos^2\theta_C = \left(\frac{\mu_2}{\mu_1}\right)^2$

$1 - (\hat{n}.\hat{p})^2 = \left(\frac{\mu_2}{\mu_1}\right)^2$

$1 - \left[\frac{4}{\sqrt{25}}\right]^2 = \left(\frac{\mu_2}{\mu_1}\right)^2 \Rightarrow \frac{\mu_2}{\mu_1} = \frac{3}{5}$

$\Rightarrow \mu_2 = \frac{3\sqrt{3}}{5}$

3. **B**

$r \propto \frac{n^2}{z}$

4. **C, D**

Stress is the property of material.

Load = stress x cross – sectional area

5. **B, C**

$(f_A + f_B)_{max}$

$2mg\sin\alpha < 2mg\cos\alpha$

$\tan\alpha < 1 \Rightarrow \alpha < 45^o$

$f_A = mg\sin\alpha + N$

$f_B = mg\sin\alpha - N$

To show $N \neq 0$

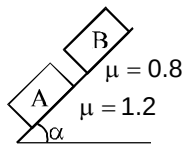
$f_{b_{max}} = 0.8\ mg\cos\alpha$

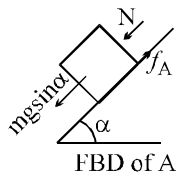
$\Rightarrow N = mg[\sin\alpha - 0.8\cos\alpha]$

$N = mg\sec\alpha[\tan\alpha - 0.8]$

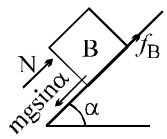
For $\alpha > \tan^{-1}(0.8)$

$f_A > f_B$ for $\alpha \leq \tan^{-1}(0.8)\ f_A = f_B$





FBD of A



6. **A, B, D**

Impulse exist in action reaction pair during collision so magnitude will remain the same.

(a) Elastic collision principle

(b) Internal force

(d) If $m_1 >>> m_2$

Direction of velocity of motion does not change

$\vec{v}_1 = (1+e)\vec{v}_c = e\vec{v}_1 \left(\vec{v}_c \approx \vec{v}_1\right)$

$\vec{v}_1 = \vec{v}_c$

7. **C, D**

KE of disc $= \frac{1}{2}MV_C^2 + \frac{1}{2}I_c\omega^2$

$= \frac{1}{2}mv^2 + \frac{1}{2}\times\frac{1}{2}ma^2\left(\frac{v}{4a}\right)^2$

KE or rod $= \frac{1}{2}I\omega^2 = \frac{1}{2}\frac{M(3a)^2}{3}\left(\frac{v}{4a}\right)^2$

KE of system = $\frac{1}{2}I_{\text{system}}\,\omega^2$

$= \frac{1}{2}\left[\frac{ma^2}{2} + m(4a)^2 + \frac{m(3a)^2}{2}\right]\left(\frac{v}{4a}\right)^2$

8. **A, B, C, D**

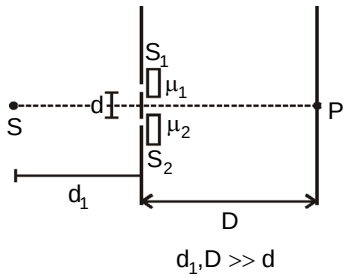
Optical path difference due to

S_1 is $= t(\mu_1 - 1) = 9\lambda\left(\frac{4}{3} - 1\right) = 3\lambda$

Optical path differ due to S_2 is $= t(\mu_2 - 1) = 2\lambda\left(\frac{3}{2} - 1\right) = \lambda$

Path diff. at point P $P = 3\lambda - \lambda = 2\lambda$

Phase diff. $\frac{2\pi}{\lambda}\times 2\lambda = 4\pi$



So net intensity at point P is

$I_{net} = I_0 + I_0 + 2I_0\cos 4\pi$

$I_{net} = 4I_0$

Optical path is conserved by S_1 is more so fringe will shift at upper side.

9. **B**

$\frac{1}{2}I_x\omega^2 + \frac{1}{2}I_y\omega^2 = \frac{f}{2}kT\left\{f_{\text{rotational}} = 2\right\}$

$I_x = I_y \Rightarrow \mu r^2 = \frac{m}{2}a^2\left\{\mu = \frac{m_1m_2}{m_1+m_2}\right\}$

$\frac{ma^2}{2}\omega^2 = \frac{2}{2}kT \quad \mu = \frac{m}{2}$

$\omega = \sqrt{\frac{2kT}{ma^2}}$

10. **B**

SECTION – C

11. **2**

Applying conservation of energy

$$mgs(\sin\alpha - \sin\beta) = \frac{1}{2}I\omega^2 + 2\left(\frac{1}{2}mv^2 + \frac{1}{2}I\omega^2\right)$$

where $\omega = v/r$ & $I = mr^2/2$

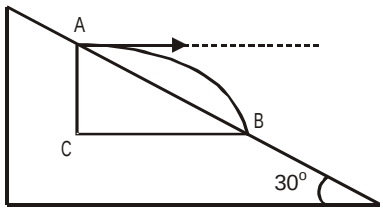
Putting values & solving

$$mgs(\sin\alpha - \sin\beta) = \frac{7}{4}mv^2$$

$$\therefore v = 2\sqrt{\frac{1}{7}gs(\sin\alpha - \sin\beta)}$$

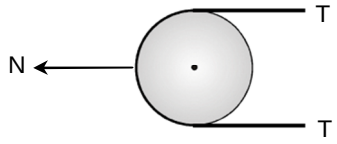
$$= 2\sqrt{\frac{10\times 3.5}{7}\left(\frac{4}{5} - \frac{3}{5}\right)} = 2 \text{ m/s.}$$

12. **5**



$$AC = \frac{1}{2}\times 10\times\left(\frac{1}{\sqrt{2}}\right)^2 = 2.5\text{m} \Rightarrow AB = 5 \text{ m}$$

13. **8**



$$2T - N = m\omega^2 R_{cm} = \mu\times\pi R\times\left(\frac{V}{R}\right)^2\times\frac{2R}{\pi}$$

$$2T - N = 2\mu v^2$$

$$N = 2T - 2\mu v^2 = 2\times 10 - 2\times 1\times 4 = 12 \text{ N}$$

14. **1**

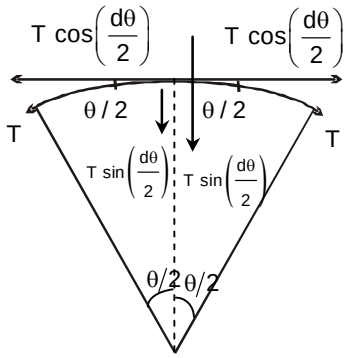
$$v = \sqrt{\frac{k(\Delta x)2\pi R}{m_o \ell}};\ k\Delta x = T$$

$$2T\sin\left(\frac{d\theta}{2}\right) = (dm)\frac{v^2}{R}$$

$$\text{or } T(2\pi) = m\frac{v^2}{R}$$

$$(k\Delta x)2\pi = (m_0\ell)\frac{v^2}{R}$$

$$v = \sqrt{\frac{k(\Delta x)2\pi R}{m_0\ell}} \ \& \ \Delta x = (2\pi r - l)$$



15. **3**

$\Delta\vec{p}$ (change in momentum per collision)

$= 2m\sqrt{2gh} = 2\times15\times10^{-3}(10) = 0.3$ kg ms^{-1}

Then in Δp_{100} (in 1 second) $= 30$ kg ms^{-1}.

Force exerted = 30 N

So weight required = 30 N. So mass = 3 kg

16. **4**

Force constant k = $\frac{100}{0.1}10^4\frac{N}{m}$

$W = \frac{1}{2}k\left(x^2_{final} - x^2_{initial}\right) = 4$ J

17. **1**

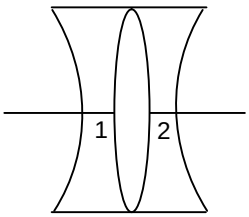
$\omega = \frac{\omega_o}{1-a\theta/r_o}, T = mr_o\omega_o\theta$

18. **4**

The Half of the maximum current is equal to $i_1 = \frac{B\omega l^{\ 2}}{4R}$ at t = $\frac{L}{R}$In2

The torque at this instant $= \frac{B^2\omega l^{\ 4}}{8R} = 4N-m$

19. **5**



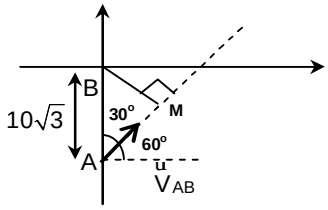
20. **6**

$\omega = \frac{1}{\sqrt{LC_{eq}}} = 100\pi$

$\Rightarrow T = 1/50$ sec

SECTION – D

21. **00003.00**



As observed by B motion of A is along AM and BM is the shortest distance between them. Relative displacement of A w.r.t to B is AM = AB cos 30° $\vec{V}_{AB} = 5\hat{i} + 5\sqrt{3}\,\hat{j}$

Time taken $(t) = \left|\frac{\vec{S}_{AB}}{\vec{V}_{AB}}\right| = \frac{AM}{V_{AB}} = \frac{10\sqrt{3}\cos 30^o}{10} = 1.5\,\text{sec}$

22. **00003.00**

$L_1 = \frac{\mu_o N_1^2 A}{l}, L_2 = \frac{\mu_o N_2^2 A \mu_r}{l}$

Since $V = -L\frac{dI}{dt}$, So

$\frac{V_1}{V_2} = \frac{L_1}{L_2} = \left(\frac{N_1}{N_2}\right)^2 \frac{1}{\mu_r} = \left(\frac{600}{200}\right)^2 \times \frac{1}{3} = 3$

23. **01000.00**

$\frac{dL}{dt} = \frac{dm}{dt}(v_1 - v_2)R = 200 \times (5 - 2.5) \times 2 = 1000\ \text{J}$

Chemistry

PART – II

SECTION – A

24. **D**

A & B are peroxides & hydroperoxides which oxidizes Fe^{2+} to Fe^{3+}

25. **A**

Let us take 100 gm solution $\Rightarrow$ 10 gm H_2SO_4 in 90 gm H_2O

$\frac{100}{1.1}$ ml solution $= \frac{10}{98}$ mol. H_2SO_4

So, $M = \frac{10 \times 1.1}{98 \times 0.1} = \frac{110}{98} = 1.122 \Rightarrow N = 2.244$

$pH = -\log 2.244$

26. **A**

Use concept of lattice energy & hydration energy.

27. **B, C**

$E^{o}_{cl^- / AgCl/Ag} = E^{o}_{Ag^+/Ag} + \frac{RT}{F} \ln K_{sp}$ .....(i)

$E^{o}_{Br^-}$ / AgBr / Ag $= E^{o}_{Ag^+/Ag} + \frac{RT}{F} \ln K_{sp}$ .....(ii)

as $E^{o}_{Ag^+/Ag}$ will be cancelled on subtractions (i) & (ii)

So E_{cell} can be determined

28. **B, C, D**

A is PhNHOH, B is PhNO

$PhNHOH \xrightarrow{T.R} Ag + PhNO$

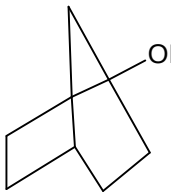
29. **B, D**

Thermal stability $HOCl < HClO_2 < HClO_3 < HClO_4$

B.P., $BiH_3 > SbH_3 > \underset{\Downarrow}{NH_3} > AsH_3 > PH_3$

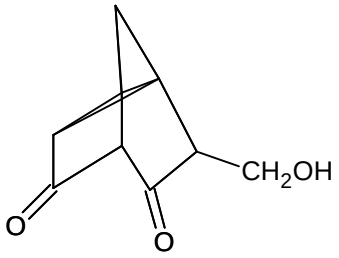
has hydrogen bonding

30. **B, C**

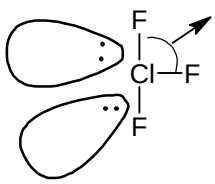
(A) 

& EtI obtained

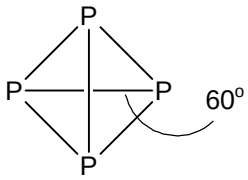
(C) HI is anhyd. So carbocation is not formed

(D)  is formed

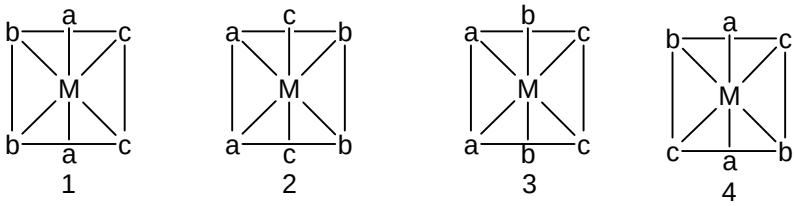
31. **A, B, C, D**



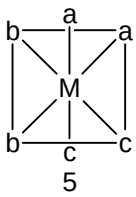
Less than 90° due to lP - BP repulsion



32. **C**



oppically inactive



only optically active

So str. (5) & its enantiomer is optically active.

33. **A**

Mabcd (tehahedral) can not show G.I & Ma_3b_3 & Ma_4b_2 can not show optical activity.

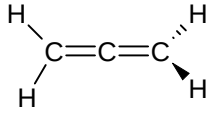
SECTION – C

34. **3**

Picric acid, HCl, P-nitro benzoic acid are more acidic than formic acid.

35. **7**

In PCl_5, maximum four atoms are in same plane



- max five atoms are in same plane $\left[N \equiv C - Ag - C \equiv N\right]^- -$ five atoms are in same plane

36. **6**

Refer EAN rule

$36 = At.No. - 2 + 6 \times 2$

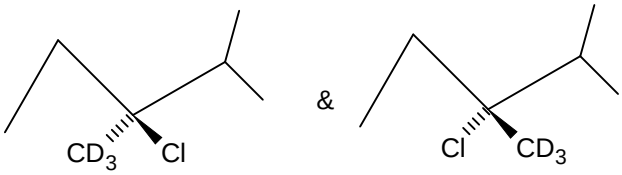
$At.No = 26$

$1S^2, 2S^2, 2P^6, 3S^2, 3P^6, 4S^2, 3d^6 \Rightarrow M^{2+} \Rightarrow 3d^6$

37. **5**

a, b can be H, H or H, CH_3 or CH_3, H or CH_3, CH_3 or H, C_2H_5 respectively.

38. **1**



39. **3**

$H_2O_2 \longrightarrow H_2O + O_2$

$NaOCl + H_2O_2 \longrightarrow O_2 + NaCl + H_2O$

In 1 min 0.02 mol/lt H_2O_2 decomposes so in 20 min

$k = \frac{2.303}{t}\log\frac{a}{a-x}$

$0.02 = \frac{2.303}{23.03}\log\frac{0.5}{(a-x)}$

$0.2 = \log\frac{0.5}{a-x}$ (take log 1.5 = 0.2)

$1.5 = \frac{0.5}{a-x}$

$a - x = \frac{0.5}{1.5} = \frac{1}{3}$

$\frac{1}{3}$mole H_2O_2 is left so $\frac{1}{3}$mole O_2 is produced in second reaction.

40. **8**

Meq of Fe^{2+} = meq of $KMnO_4$ = $\frac{8 \times V \times 1 \times 1000}{100 \times 56} = N_{KMnO_4} \times V$

$N = \frac{8}{5.6}$, as eq. wt. of $KMnO_4$ is same in both experiment so N_{KMnO_4} is same in both

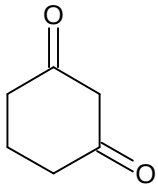
Meq of $KMnO_4$ = meq of H_2O_2

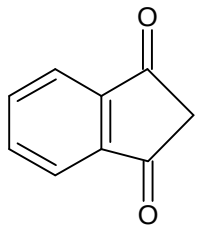
$$20\times\frac{8}{5.6}=20\times N=20\times\frac{\text{vol.strength}}{5.6}$$

so vol stre. of $H_2O_2=8$

41. **4**

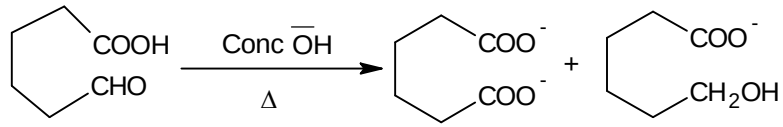
acetophenone, resorcinol,

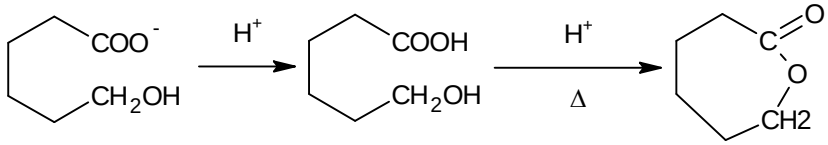




will give Iodoform test

42. **2**

X = 

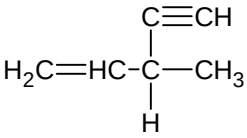


43. **3**

Given halophenols are more acidic than phenol.

SECTION – D

44. **00080.00**



45. **00017.92**

$$2CH_3COOK\xrightarrow{\text{electrolysis}}C_2H_6+2CO_2+H_2+2KOH+2H_2O$$

$Q=2\times402\times60$

moles of $e^-=0.5$; used mole of $e^-=\frac{0.5\times80}{100}=0.4$

as 2 mole $e^-\longrightarrow 4$ mole gases

so 0.4 mole $e^-\longrightarrow\frac{0.4\times4}{2}=0.8$ mole gases

$V=0.8\times22.4$ Lt. = 17.92 Lt.

46. **00049.50**

$M_{hypo} = \frac{1.24 \times 1000}{248 \times 40} = 0.125$

meq of hypo = meq of I_2 in 25 ml $= 0.125 \times 1 \times 40 = 5$

meq of I_2 = meq of $As_2O_3 = 5$

meq of AS_2O_3 in 20 ml solution = 5

so meq of AS_2O_3 in 400 ml = $\frac{5 \times 400}{20} = 100$

mole of $AS_2O_3 = \frac{100 \times 10^{-3}}{2} = 50 \times 10^{-3} = 0.05$

wt. of $AS_2O_3 = 0.05 \times 198 = 9.9$ g

wt.% $= \frac{9.9}{20} \times 100 = 49.5\%$

Mathematics

PART – III

SECTION – A

47. **C**

$\because \alpha = \dfrac{4}{x_1}$ and $\beta = \dfrac{-3}{y_1} = \dfrac{-4}{\sqrt{x_1^2 - 4}}$

$\therefore \displaystyle\int_2^{4\sqrt{2}} \frac{4/x_1}{-4/\sqrt{x_1^2-4}}\,dx_1 = -\int_2^{4\sqrt{2}} \frac{dx_1}{x_1\sqrt{x_1^2-4}}$ $\quad (\text{Put } x_1 = 2\sec\theta,\ dx_1 = 2\sec\theta\tan\theta d\theta)$

$= -\displaystyle\int_0^{\frac{\pi}{4}} \frac{2\sec\theta\tan\theta d\theta}{2\sec\theta.2\tan\theta} = -\frac{\pi}{8}$

48. **B**

$\because \displaystyle\int_0^x \frac{t^3}{1+3t^2}dt + \int_0^x \frac{tdt}{1+3t^2} = f(x)$

Put $1+3t^2 = u^2 \Rightarrow$

$= \displaystyle\int \frac{\frac{1}{3}udu \times \left(\frac{u^2-1}{3}\right)}{u^2} + \frac{1}{6}\ln\left(1+3t^2\right)\Big|_0^x$

$= \frac{1}{9}\displaystyle\int_1^{\sqrt{1+3x^2}} \left(u - \frac{1}{u}\right)du + \frac{1}{6}\ln\left(1+3x^2\right)$

$= \frac{1}{9}\left(\frac{u^2}{2} - \ln u\right)_1^{\sqrt{1+3x^2}} + \frac{1}{6}\ln\left(1+3x^2\right)$

$= \frac{1}{18}\left(\left(1+3x^2\right)-1\right) - \frac{1}{9}\ln\sqrt{1+3x^2} + \frac{1}{6}\ln\left(1+3x^2\right)$

$= \frac{1}{6}x^2 + \left(\frac{-1}{18} + \frac{1}{6}\right)\ln\left(1+3x^2\right)$

$= \frac{1}{6}x^2 - \frac{1}{9}\ln\left(1+3x^2\right)$

49. **A**

$x\left(2x^2y^3+1\right)dy = y\left(1-3x^2y^3\right)dx$

$\Rightarrow x^2y\left(2xdy + 3ydx\right) = \dfrac{ydx - xdy}{y^2}$

$\Rightarrow d\left(x^3y^2\right) = d\left(\dfrac{x}{y}\right)$

$\Rightarrow x^3y^2 = \dfrac{x}{y} + c$

$\Rightarrow x^3y^3 = x + cy$

50. **A, B, C, D**

$$x = \frac{12}{15} \times \frac{1}{\left(1 + \frac{1}{x^2 + y^2}\right)} \quad \ldots\ldots\ldots \text{(i)}$$

$$\text{and } y = \frac{4}{5} \times \left(\frac{1}{1 - \frac{1}{x^2 + y^2}} \right) \quad \ldots\ldots\ldots \text{(ii)}$$

Put $x^2 + y^2 = \frac{1}{u}$ and add (i)^2 and (ii)^2

$$\Rightarrow \frac{1}{u} = \left(\frac{4}{5}\right)^2 \left(\frac{9}{(1+4)^2} + \frac{1}{(1-4)^2} \right)$$

$$\Rightarrow 25\left(u^2 - 2 + \frac{1}{u^2}\right) = 32\left\{5\left(4 + \frac{1}{4}\right) - 8\right\}$$

Put $u + \frac{1}{u} = v \Rightarrow v = \frac{6}{5}$ or $\frac{26}{5}$

$$\Rightarrow 4 = 5 \text{ or } \frac{1}{5}$$

$$\Rightarrow (x, y) = (2, 1) \text{ or } \left(\frac{2}{5}, \frac{-1}{5}\right)$$

51. **A, B, D**

$\because \Delta SAB$ is right angle at S

Centre C is $\left(\frac{1+7}{2}, \frac{1-6}{2}\right) = (4, -3)$

$\therefore \alpha = 4$ and $\beta = -3$

$\therefore \alpha + \beta = 1$ and $\alpha - \beta = 7$

SC is radius and AB is diameter

$$\therefore SC = \frac{1}{2}\sqrt{36 + 49} = \frac{1}{2}\sqrt{85} = \sqrt{\frac{85}{4}} = \sqrt{21.25}$$

52. **A, C**

$\because \quad x \to 0^- \Rightarrow \{x\} = 1 + x$

$$\therefore \quad \lim_{x \to 0^-} f(x) = \lim_{x \to 0} \frac{\sin^{-1}(-x)\cos^{-1}(-x)}{\sqrt{2(1+x)}.(-x)} = \frac{\pi}{2\sqrt{2}}$$

& for $x \to 0^+ \Rightarrow \{x\} = x$

$$\lim_{x \to 0^+} f(x) = \lim_{x \to 0} \frac{\sin^{-1}(1-x)\cos^{-1}(1-x)}{\sqrt{2x}(1-x)} = \frac{\pi}{2}$$

53. **B, D**

$\because \quad \int_0^{\pi} \sin x dx = 2$

$\Rightarrow \quad b - a = 4\pi$

$$\int_0^{a+b} \cos x dx = 9$$

$\Rightarrow \quad a+b=\frac{9\pi}{2}$

$\therefore \quad a=\frac{\pi}{4} \& b=\frac{7\pi}{4}$

54. **B, C**

$\because \quad ab+bc+ca \le \left(a^2+b^2+c^2\right)$ for all a, b, c $\in R$

$\Rightarrow \quad ab+bc+ca \le 6$

Again, $a^2 > b^2+c^2-2bc \qquad (\because \cos A < 1)$

$\therefore \quad \frac{a^2+b^2+c^2}{2} < (ab+bc+ca)$

$\Rightarrow \quad ab + b + ca > 3 \qquad$ So, (B) & (D)

55. **C**

56. **B**

Locus of Q is an ellipse whose vertices z1 and z2 don't lie on it and eccentricity is $\sqrt{\frac{2}{3}}$ for locus of Q.

Now, $2ae = 2\times\sqrt{\frac{2}{3}}$

$|z_1 - z_2| = 2$

$\therefore [m] = 2 \ (\because m < |z_1 - z_2|)$

SECTION – C

57. **5**

$\because \quad k = \int_0^2 f(x)dx + \int_2^4 f(x)dx - 4$

$= \int_0^2 f(x)dx + \int_0^2 f(x+2)dx - 4$

$= \int_0^2 2.dx - 4 = 0$

& $\quad \alpha = \beta = \frac{1}{1} \Rightarrow q = 1$

$\therefore \quad$ roots of $g(x) = ax^2 - bx + c$ lying in (0, 1)

$\Rightarrow \quad g(0).g(1) = C(a-b+c) \in N \qquad$ so least value is "1"

$= a^2\alpha_1\beta_1(1-\alpha_1)(1-\beta_1) \qquad$ as α_1, β_1 are roots of $ax^2 - bx + c = 0$

$g(0)g(1) < \frac{a^2}{16} \qquad ax^2 - bx + c = 0$

$\Rightarrow \quad \frac{a^2}{16} > 1 \qquad \because \quad \alpha_1(1-\alpha_1) \le \frac{1}{4}$

$\Rightarrow \quad a > 4 \quad$ or $\quad a < -4 \quad \& \quad \alpha_1 \ne \beta_1$

$\Rightarrow \quad$ least value of a = 5

58. **6**

$$f(n) = {}^{2n}c_2 - n = \frac{2n(2n-1)}{2} - n$$

$$= n(2n-1-1) = 2n(n-1)$$

$$\frac{1}{f(n)} = \frac{1}{2}.\frac{1}{n(n-1)} = \frac{1}{2}\left[\frac{1}{n-1} - \frac{1}{n}\right]$$

$$\frac{1}{f(4)} = \frac{1}{2}\left[\frac{1}{3} - \frac{1}{4}\right]$$

$$\Rightarrow \quad \sum_{n=4}^{\infty} \frac{1}{f(n)} = \frac{1}{2}\left(\frac{1}{3}\right) = \frac{1}{6}$$

59. **4**

$\because \quad f(x) = x + \lambda.x^2 + \mu.x$

where $\lambda = \int_0^1 zf(z)dz \;\&\; M = \int_0^1 z^2 f(z)dz$

then

$$\lambda = \frac{80}{119} \qquad \& \qquad \mu = \frac{61}{119}$$

$$\therefore \quad f(x) = \frac{20}{119}\left(4x^2 + 9x\right)$$

$$\Rightarrow \quad \lim_{x\to\infty} \frac{f(x)}{ax^2 + 7x} = \frac{\frac{20}{119} \times 4}{a} = \frac{20}{119}$$

$$\Rightarrow \quad a = 4$$

60. **3**

$$\because \quad x^2\left(x^2 + \frac{1}{x^2}\right) = 7\left(x^2 + 2x + 1\right)^2$$

$$x^2 + \frac{1}{x^2} = 7\left(x + 2 + \frac{1}{x}\right)^2$$

Let $\quad \alpha + \frac{1}{\alpha} = t$

$$\Rightarrow \quad t^2 - 2 = 7(t+2)^2$$

$$\Rightarrow \quad t = -3 \quad \text{or} \quad -\frac{5}{3}$$

$$\Rightarrow \quad |t| = 3 \quad \text{or} \quad \frac{5}{3}$$

61. **8**

$\because \quad x^2 = -4(y-1)$ has directrix

$y - 1 + 1$

$y = 2$

$\therefore \quad \lambda = 4$

$\mu = -4$

$\lambda - \mu = 8$

62. **2**

$\Delta_1 = 9 - 4a \quad \Rightarrow \quad \Delta_1 + \Delta_2 = a^2 - 4a + 5$

$\Delta_2 = a^2 - 4 \qquad\qquad = (a-2)^2 + 1 > 0$

63. **3**

$\therefore \quad z_1 = z \Rightarrow z_2^2 = z^2\omega^2, z_3 = z_2^2\omega^2$

$\therefore \quad \frac{z_2}{z_1} = +\omega \quad \text{or} \quad -\omega$

$\frac{z_3}{z_1} = +\omega^2 \quad \text{or} \quad -\omega^2$

64. **3**

$\because$ Plane is $\frac{x}{a} + \frac{y}{b} + \frac{z}{c} = 1$

Distance from (0, 0, 0) = 3 is

$$\frac{1}{\sqrt{\frac{1}{a^2} + \frac{1}{b^2} + \frac{1}{c^2}}} = 3$$

$\therefore$ Locus of P is $\frac{1}{x^2} + \frac{1}{y^2} + \frac{1}{z^2} = \frac{1}{9}$

65. **5**

Let $\bar{a} = x\bar{b} + y\bar{c} + z(\bar{b} \times \bar{c})$

$\Rightarrow \bar{a}.\bar{b} = x(\bar{b}.\bar{b}) + y(\bar{b}.\bar{c}) = 2$

$x + y(\bar{b}.\bar{c}) = 2$ ..... (1)

$\bar{a}.\bar{c} = 0 + y \times 1 = 5$

$y = 5$

$\bar{a}.(\bar{b} \times \bar{c}) = 0 + 0 + z|\bar{b} \times \bar{c}|^2$

$z = 2$

66. **4**

$\because \tan^{-1} x \geq \frac{2\pi}{3} - \frac{\pi}{2}$ for $-1 \leq 2 - x^2 \leq 1$

$\Rightarrow \tan^{-1} x \geq \frac{\pi}{6} \qquad -3 \leq -x^2 \leq -1$

$x \geq \frac{1}{\sqrt{3}} \qquad 1 \leq |x| \leq \sqrt{3}$

$\therefore 1 \leq x \leq \sqrt{3}$

$\therefore$ have no solution for $x < 1$ or $x > \sqrt{3}$

So, $\alpha_{max} = 1$ and $\beta_{min} = \sqrt{3}$

SECTION – D

67. **00001.12**

$\because \dfrac{2x}{\sqrt{x^2+1}} + \dfrac{\sqrt{x^2+1}}{x} \geq 2\sqrt{\dfrac{2x}{\sqrt{x^2+1}} \times \dfrac{\sqrt{x^2+1}}{x}} = 2\sqrt{2}$

$\therefore M = 2\sqrt{2}$ & $m = \pm 1$, i.e., $m = 1$

68. **00000.75**

Put $t = \tan\theta$

$\Rightarrow \displaystyle\int_0^{\pi/2} \frac{\sec^2\theta d\theta}{\sec^2\theta\left(1+\tan^{2017}\theta\right)} = \frac{\pi}{4}$

$\Rightarrow \dfrac{3x}{\pi} = \dfrac{3}{4} = 0.75$

69. **00000.75**

Let $t = \log_{(3x+4)}(2x+1)$

$\Rightarrow \quad 2t + \dfrac{1}{t} = 3 \quad \Rightarrow \quad t = 1 \text{ or } \dfrac{1}{2}$

$\log_{(3x+4)}(2x+1) = \dfrac{1}{2} \Rightarrow x = -1 \text{ or } \dfrac{3}{4}$

$\therefore$ only solution is $x = \dfrac{3}{4}$